

Lab 3 – Architectural Pattern Selection

Scenario: Self-Service Coffee Kiosk System

Architecture Selection

We chose **Layered Architecture** for the Self-Service Coffee Kiosk System.

Architectural Choice and Justification

Layered Architecture is suitable for this system because it divides the application into separate layers such as the **Presentation Layer, Business Layer, and Data Layer**. This provides a clear separation of responsibilities and makes the system easier to understand and maintain. The lab handout specifically describes layered architecture as appropriate for traditional applications because of its clear separation of concerns. Lab_3_Architecture_Student_hand...

The first reason for selecting Layered Architecture is that the kiosk contains different types of functionality. The **User Interface Component** handles touchscreen interaction, while the **Order Manager Component** handles ordering logic. The **Payment Service Component** processes credit-card payments, the **Receipt Printer Component** handles receipt printing, and the **Database Component** stores menu and pricing information. Separating these responsibilities makes the design more organized. The second reason is that the Self-Service Coffee Kiosk is a relatively small system with a simple and predictable workflow. A customer selects a coffee and size, places the order, completes payment, and receives a receipt. Therefore, a layered architecture is simpler and more suitable than a more complex architecture such as Microservices.

Security Advantage

A security advantage of this architecture is that payment-related operations can be handled separately by the **Payment Service Component** instead of being directly handled by the user interface. This helps isolate payment functionality from the other parts of the system and limits the components that need to interact with payment processing.

Performance Benefit

A performance benefit of Layered Architecture for this kiosk is its simple communication flow. User requests move from the **User Interface** to the **Order Manager**, which communicates only with the required components such as the Payment Service, Receipt Printer, or Database. Since the kiosk has a limited set of functions, this provides a straightforward and efficient design without the extra network and operational complexity associated with Microservices Architecture. Lab_3_Architecture_Student_hand...

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